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CS-499 Computer Science Capstone 2025 C-5 (Sep - Oct)

5-2 Milestone Four: Enhancement Three: Databases

The artifact I selected for this enhancement is a travel web application built with Angular (frontend), Node.js/Express (backend), and MongoDB (database). It was originally created during CS-465: Full Stack Development. I chose to include this artifact in my ePortfolio because it highlights my ability to design, implement, and test a functional full-stack application.

For this milestone, I enhanced the artifact by improving the way the application interacts with the database through a new filtering and sorting feature. I implemented a new RESTful API route (/api/trips/filter) in the backend that allows users to search and sort trip data directly through MongoDB queries. This feature uses Mongoose's find() and sort() methods to efficiently handle query parameters for filtering trips by destination and sorting them by price or date. By handling these operations at the database level instead of the frontend, the application became faster, more efficient, and scalable.

I selected this artifact because it highlights my skills in database design and integration. The enhancement demonstrates how I can build flexible and optimized query logic using MongoDB and Mongoose, and how to connect it to a dynamic Angular interface.

This enhancement aligns with the database outcome by demonstrating my ability to use innovative techniques for managing and accessing data. It also shows my understanding of how

to design backend logic that supports user-driven data manipulation without compromising performance or security.

During this process, I learned how to write more efficient database queries and how to structure routes and controllers to handle dynamic parameters. I also learned the importance of validating query inputs and providing clear responses for cases where no results are found. One challenge I faced was ensuring that the filtering logic worked correctly with both destination searches and sorting options, but I was able to overcome it by testing different combinations of query parameters and validating them through Postman and the Angular interface.

Enhancing this artifact helped me strengthen my skills in backend development and database integration. It taught me how to extend the functionality of an existing full-stack system while maintaining data integrity, efficiency, and usability.